

The Data

Earl Wong

Motivation

In the past, algorithms were developed using a combination of:

- 1) Fundamental theory and the understanding of physics
- 2) Mathematical tools and models (biological, language, etc.)
- 3) Physical, human intuition
- 4) Heuristics / simplifications

Motivation

- In the deep learning era, 1) and 2) are still of great importance.
- In the deep learning era, the information contained in 1) and 2) are now extracted / learned from a training dataset / input tensors.
- In the deep learning era, the needed information is captured in a CNN and / or Transformer model via back propagation (learning signal / error signal).

Motivation

The key components:

- 1) The data used to learn the knowledge.
- 2) The model used to “collect / store” the knowledge.
- 2) The error signal, allowing the model to acquire the knowledge.

Motivation

- Regarding 2): In 2025, it's transformers and cnn's.
- Regarding 2): Successful variants / combinations include Vision Transformers (ViT), U-Nets with skip connections, etc.
- Regarding 3): There exists a plethora of loss functions for different objectives - cross entropy, mse, regularization, etc.
- Regarding 2 & 3): There exists different “tricks” (skip connections, normalization, etc.) to expedite rapid and successful learning.

Motivation

- Regardless of whether we are learning 1) a large language model, 2) an imaging or computer vision model, 3) a combination of both (VLM), or, 4) a vision, language, and action model (VLA), learning is achieved from the data.
- Although loss functions and models are also very important, in this presentation, we will focus on the data.
- Specifically, we will focus on the data for imaging and computer vision problems.

Outline

- We will start with the data for computer vision problems.
- We then will migrate to the data for models like CLIP, and, its associated shortcomings.
- We will then discuss DINO and the data / model architecture phenomena.
- We will then drill down into the data for a specific imaging problem - denoising.

Computer Vision

- Classification, detection, and segmentation experienced a huge jump in quality, with the advent of deep learning and labeled training data.
- Classification leveraged ImageNet, Detection leveraged MS COCO, while Segmentation leveraged CityScapes and ADE20K.
- ImageNet was used to pre-train a CNN model backbone for classification.
- Fine tuning for Detection and Segmentation then occurred, using smaller, task specific datasets.
- ***Finally, for productization, proprietary datasets are introduced to make the end product “shippable / robust”.

Computer Vision

- Initially, image classification via deep learning and ImageNet was somewhat brittle.
- Performance significantly improved with data augmentation - image crop, rotation, flip, color shift, etc.
- As mentioned in an earlier presentation, dataset imbalance was also a very serious problem, that hindered performance.
- In order to cut to the chase, let's assume that we have executed the above correctly, **achieving high benchmark performance.**
- Are we done? NO!

The Academia | Industry Divide

- Historically, academia would create / innovate many of the baseline algorithms.
- Then, industry would evaluate their real world performance.
- If the performance was good, industry would make the algorithm deployable - make the algorithm run faster, use a smaller memory footprint, introduce simplifying heuristics, etc.
- Over the course of time, tried and true shipping algorithms emerged - Burt & Adelson Laplacian Pyramids, David Lowe's SIFT, Harris Corner Detector, etc.

The Academia | Industry Divide

- In the deep learning era, a similar duality exists.
- Only now, the baseline model determines the benchmark result.
- Industry then needs to supplement the training data, enriching the baseline model to make the model shippable.

The Academia | Industry Divide

Ideally, the approach should be:

- 1) Evaluate the benchmark dataset for “coverage”.
- i.e. Look through the dataset, image by image, or video sequence by video sequence, to see “what is represented”.
- 2) Create a new “shipping criteria” test set, with real world deployment use cases, edge cases, etc. that address “what is missing”.

The Academia | Industry Divide

- 3) Evaluate the learned model on the shipping criteria dataset.
- 4) Collect the data needed to retrain / address the shortcomings / gaps / corner cases, not represented in the learned model.
- Note: The later a shipping problem is discovered, the greater the \$\$\$\$\$\$.

The Academia | Industry Divide

- Q: How do you determine what is missing?
- A: You have to “live the problem”.
- Analogy: Is statistics just a bunch of equations, or, do you see statistics everywhere around you?
- If the latter is true, you are “living the problem”.
- The same holds for images and video.

Computer Vision

Let's look at a practical problem, with the “dataset gap issue” in mind. Would you “trust” a CNN based pedestrian detector for self driving vehicles:

- If the detector was only trained on frontal and partial side images of a pedestrian?
- If the detector was only trained on pedestrians that were standing?
- If most of the dataset contained images of pedestrians in an indoor environment?
- If the detector was only trained on pedestrians captured in low dynamic range scenes / only cloudy “Rochester, NY” days?
- If the detector was only trained on images where the height of the pedestrian occupied between 20%-80% of the vertical pixels?

Computer Vision

- This is the messiness of the real world ... a messiness that is not be fully represented in a benchmark dataset.
- Note: We have only scratched the surface of the potential problems, by listing the OBVIOUS shortcomings.
- Hence, collecting additional data beyond the benchmark is an absolute necessity for creating a shippable product.

Remember

- The benchmark is not targeted for shipping.
- The benchmark is targeted for baseline performance.

Computer Vision

Remember / recall:

- Lena, Peppers, Boat, etc. were academic evaluation metrics for non-deep learning algorithms.
- One would not ship, just because of great performance on Lena, Peppers, Boat, etc.
- Benchmarks are the modern day academic evaluation metric for deep learning.
- Benchmarks are a necessary criteria for shipping, **BUT**, benchmarks are NOT A SUFFICIENT criteria for shipping.
- Finally, in the deep learning era, deep learning models only know what they have been trained on / what is in the training data.

Computer Vision

- REMINDER1: Suppose you take any ImageNet object category, and capture an image of the object at a “strong angle / from a non-frontal viewpoint”.
- Will it work? Why or why not?
- What is the problem with the learned representations?
- REMINDER2: Suppose you input the following image into your deep learning classifier:
- Will you obtain the correct classification?
- Why or why not?



Computer Vision

Takeaways

- 1) The benchmark is the beginning / the benchmark is meant for academic pursuit.
- 1) The benchmark ONLY means that you have achieved a necessary threshold for industry evaluation.
- 2) “Whack a mole” is a bad approach.
- 2) “Seeing / knowing” **what is missing** from the benchmark will save you \$\$\$\$\$.
- 1) and 2) Your trained model is only as good as the training data it was trained on.

CLIP and DINO

- Labeled data is expensive.
- CLIP and DINO provide nice workarounds to this problem.
- CLIP was trained on captioned images scraped from the internet. (WIT - Web Image Text)
- As a result, it has a unique set of dataset bias problems.
- DINO was initially trained on ImageNet data.
- DINO is a fascinating study, from a data / model perspective.

CLIP

- CLIP moves deep learning beyond the initial classification, detection, segmentation framework - i.e. car, bicycle, etc.
- Now, we have “A picture of a boy riding a bicycle in a city park on a sunny day”.
- i.e. The problem complexity has increased multifold: there is a language model component, a computer vision component, and an alignment component.
- On top of this, there are biases in the caption data and the image data.
- Focusing on the image data biases - different ISP tunings from different camera manufacturers can adversely affect CLIP results since the learned mid level representations may become “no longer representative”.

DINO

- DINO is fascinating from a data / model perspective.
- Using ImageNet data, ResNet-50 Supervised significantly outperformed ResNet-50 DINO (self supervised)
- However, also using the same ImageNet data, ViT Supervised performed about the same as ViT DINO (self supervised).
- i.e. The transformer model significantly impacted performance.
- Note / Hint / Insight: Because of the self attention mechanism in transformers, DINO also naturally produces segmentation maps, sans segmentation training.

Information Extraction from the Data

Same data:

- CNN's are predicated on growing receptive fields, translation invariance, and hierarchical representations.
- => CNN's make the data texture centric.
- Transformers are predicated on attention, with no inductive biases.
- => Transformers make the data object centric.
- What viewpoint is your deep learning problem more aligned with?

Imaging

- The latter question provides a nice segue into our final data topic.
- Upscaling, denoising, restoration, etc. are classical imaging problems.

We focus our attention on ML de-noising, since

- 1) The hallucination problem becomes irrelevant, if residual noise is predicted.
- 2) In cell phone cameras, noise is a bigger problem than blur.

Denoising Dataset

Let's consider two benchmark datasets - Darmstadt Noise Dataset (DND) and Smartphone Image Denoising Dataset (SIDDD).

- (DND) Benchmarking Denoising Algorithms with Real Photographs, Plotz and Roth, CVPR 2017
- (SIDDD) A High Quality Denoising Dataset for Smartphone Cameras, Abdelhamed, et. al., CVPR 2018

DND provided a significant improvement versus “take a noise free image and adding gaussian noise” since

- 1) The noise free image may not be truly noise free.
- 2) True camera noise is not solely gaussian.

Denoising Dataset

Specifically, the authors of DND captured real camera data and performed the following post processing corrections

- 1) Aligned the low ISO and high ISO image captures - assuming a global translation.
- 2) Corrected for intensity scaling errors (resulting from exposure and analog gain inaccuracies) when making a low ISO capture followed by a high ISO capture.
- 3) Accounted for clipping.
- 4) Removed low frequency bias.

Denoising Dataset

- 50 scenes were captured, using cameras ranging from cell phones to full frame DSLR's.
- 33 of the 50 scenes are shown below.

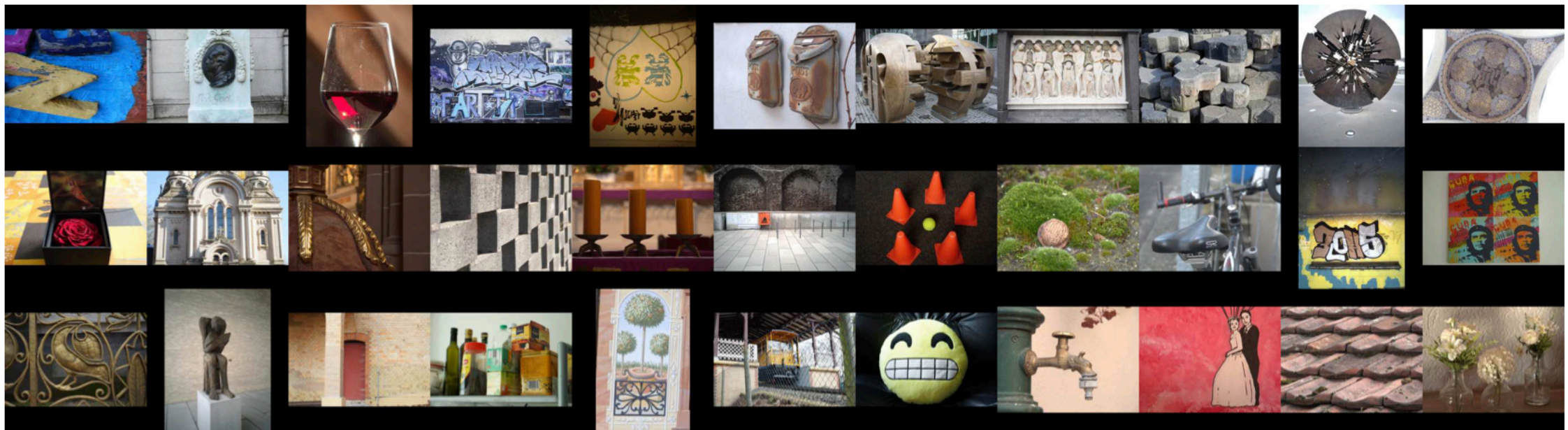


Figure 2. An overview of the scenes used in our benchmark dataset (subset shown).

Denoising Dataset

- SIDD focused the denoising dataset to solely cell phone camera captures - Apple iPhone, Google Pixel, Samsung Galaxy, Motorola Nexus, and LG G4 (2018)

SIDD produced an improved noise free reference image, by

- 1) Averaging a sequence of noisy images (vs performing a low ISO capture)
- 2) Performing dense local image alignment (vs performing a global translation correction)
- 3) Performing defective pixel correction, outlier removal, and intensity alignment.

Denoising Dataset

- 10 different scenes were captured under various conditions, using 5 cell phone cameras.
- 3 of the 10 scenes are shown below.

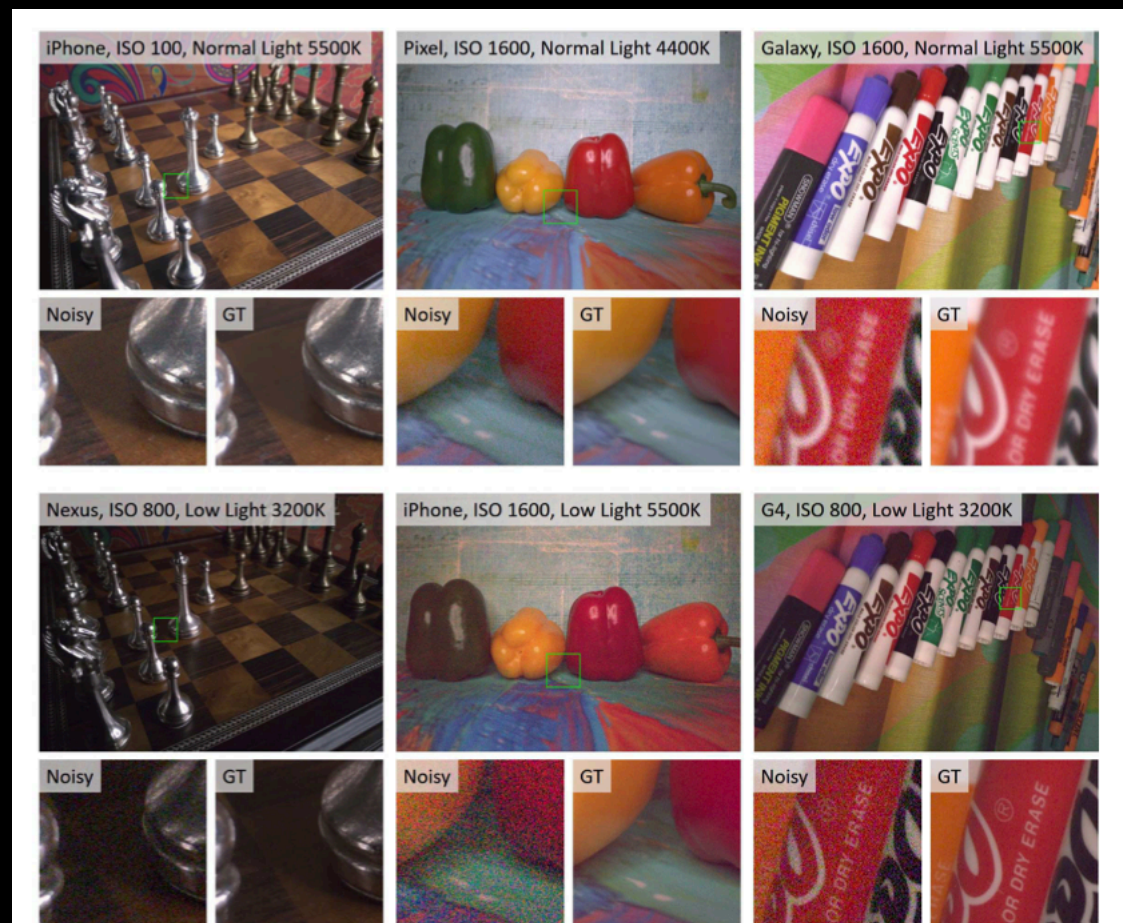


Figure 2: Examples of noisy images from our SIDD dataset captured under different lighting conditions and camera settings. Below each scene, zoomed-in regions from both the noisy image and our estimated ground truth (Section 4) are provided.

Denoising Dataset

Big result

- DnCNN performed significantly better, when trained on SIDD versus DND.
- SIDD had ground truth images that were significantly closer to a being noise free (versus DND).
- SIDD had better noise free / noisy image pair alignment (versus DND), since mobile camera OIS shifting was addressed.

Denoising Dataset

- At this point, it is probably appropriate to entertain the following questions. If your deep learning based denoising model ...
- Q1) ... performed “near perfect” on a dataset set consisting of “noise free images” and noisy images with additive gaussian noise, are you ready to ship the model on your mobile phone?
- Q2) ... performed “near perfect” on a dataset consisting of 50 real camera scenes, are you ready to ship the model on your mobile phone?
- Q3) ... performed “near perfect” on dataset consisting of 10 real camera scenes captured from 5 mobile phone cameras from 2019, are you ready to ship the model on your mobile phone?

Denoising Dataset

In addition, how far is the benchmark result from the actual shipping product?

- 1) 50 scenes (DND) and 10 scenes (SIDDD). Is this adequate for a robust evaluation? A robustly trained model?
- 2) Noise is sensor specific. Will the trained model “just generalize” to the noise produced by a 2025 Sony sensor? A 2025 Omnivision sensor?
- 3) Your deep learning model is only as good as the data it is trained on ...

Context

Let's now examine ML based denoising in a bigger context:

- Your task is XYZ = denoising.
- You want to improve the performance of XYZ.
- ML is the new tool in your bag of existing tools.
- How should you proceed?

The BIG Picture

1) Determine the current baseline noise reduction performance, under a diverse set of use cases.

- Q) What are the test vectors / use cases?
- A1) Camera settings: Illumination / exposure / ISO setting combinations
- A2) Scene Physics: Lux: 5, 25, 100 ..., Color Temperature: 2700K, 4000K, 5500K ...
- A3) Scene Types: (Flat regions) Sky, Walls, Lambertian surfaces, ... (Texture regions): Faces & Hair, Carpet, Grass, Clothing , ... (Motion): Global motion (camera pan, etc.), local motion (people, cars, bicycles moving, etc.)

The BIG Picture

2) Determine the target performance.

- Q1) Where in the pipeline, will we measure the performance?
- A1) RAW, linear RGB, after tone mapping?
- Q2) What metrics will be used to quantify performance?
- A2) PSNR or SSIM, human evaluation ...
- Q3) What is the ground truth?
- A3) Noise free image, noise free video - how do we acquire these?

The BIG Picture

3) Identify the use cases that need to be addressed.

- Q) Which scenes do we prioritize?
- A) The ones receiving the most user complaints / the ones that provide bad user experiences.
- Q) Can we / should we try to address everything / denoising using only ML?
- A) Do you want to ship anytime soon?

The BIG Picture

4) Define the Role of ML (Model)

- Q: What are the ML model details?
- A1: We will train the model using RAW data from this optical and sensor path.
- A2: We will train a model that will run on the current NPU / ANE with these runtime specs.
- A3: We will deploy a transformer or CNN model with this approximate architecture (size / depth, precision)

The BIG Picture

4) Define the Role of ML (Dataset)

- From 3), we observe the following gaps / weaknesses in our current noise reduction.
- We target this “gap” category of input images and video sequences.
- We define the following specifications for this dataset.
- Recall: Your deep learning model is only as good as the data it is trained on ...

The BIG Picture

4) Define the Role of ML (Dataset - High Level)

- Illumination (Noise)
- Scene content (Structure)
- Event (Use case / gap category)

The BIG Picture

4) Define the Role of ML (Dataset - Illumination)

a) Camera front end lock down - lens / sensor / ISO / exposure (performed for model learning)

b) Camera back end lock down (assumptions) - HDR, tone curve(s) (performed for end-to-end evaluation)

Note: Improved raw NR should translate to improved back end NR. But, you always want to make an apples to apples final output comparison.

The BIG Picture

4) Define the Role of ML (Dataset - Illumination)

- ISO / exposure settings, combinations
- Illumination range: 1 Lux to ? Lux
- Color temperature range: 2700K to 7000K
- Mixed lighting condition: tungsten and daylight, shadows and highlights

The BIG Picture

4) Define the Role of ML (Dataset - Scene Content)

a) Test pattern structure: step edges, smooth edges, ramps, lines, textures at various frequencies (focus on high frequency), Imatest charts

b) Test pattern orientations: all; Test pattern contrast: low

c) Natural scenes: sky, clouds, walls, smooth surfaces, paintings, grass, clothing / cloth, hair, other fine texture patterns, faces, objects containing text

The BIG Picture

4) Define the Role of ML (Dataset - Use case)

- Comprehensive coverage of images and video for the use case of interest.
- Use case shooting will most likely uncover yet to be discovered / new problems, because of its concentrated focus.
- Tell the shooters to shoot this, but do not tell them the end goal - noise reduction, upscaler, image classifier, image segmenter, etc.
- Illumination and Scene Content focuses on the basics of ML noise reduction - in general, what any ML noise model needs to learn.
- Use case drills down, addressing the specific gap / problem.

The BIG Picture

4) Define the Role of ML (Artifact / Regression Checks)

Input1) Corner case inputs: blurry video from focus change, fast pans in low light, etc.

Input2) Abrupt transition changes: fast illumination change, fast color temperature change, fast scene content change (capturing a new release movie in a theater), etc.

Input3) Pathological inputs that are never part of a training set: artificial test patterns, etc.

The BIG Picture

5) Consider future camera systems and future ML integration.

- Q1) What if the sensor changes, resulting in a different noise model?
- Q2) What if the sensor resolution changes, resulting in the need to process more data?
- Q3) What if binning occurs / needs to be supported?
- Q4) What if the optics change?

The BIG Picture

6) Consider next generation camera algorithms and future ML integration.

- Q5) Can we easily support 240 fps video (new feature), without significant ML hardware rework?
- Q6) What if downstream algorithms change? HDR, tone curve, sharpening, etc? Will this require a new ML dataset capture or significant rework?

=> A little bit of thought today, can save you a lot of \$\$\$\$\$\$ tomorrow.